JEE(MAIN) : ENTHUSIAST & LEADER SCORE -II

Time : 3.00 Hours **Maximum Marks : 300**

Q2

READ THE INSTRUCTIONS CAREFULLY

Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet and OMR sheet (Side-1 & Side-2) with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of 90 questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics**, **Chemistry** and **Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
   (a) Section-I contains 20 **multiple choice** questions with **only one correct** option.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
   (b) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.
   Marking scheme : +4 for correct answer, 0 if not attempted and –1 in all other cases.
5. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
6. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
7. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
8. **Do not fold or make any stray marks on the Answer Sheet.**
9. **Take g = 10 m/s^2 unless otherwise stated.**

Name of the Candidate (in Capitals) : _____

Form Number : in figures _____

: in words _____

Centre of Examination (in Capitals) : _____

Candidate's Signature : _____ **Invigilator's Signature :** _____

DO NOT BREAK THE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Topic : Full Syllabus

PART-1 : PHYSICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. Mass defect in a $_{11}Na^{23}$ atom is 0.21 amu. Then the binding energy per nucleon will be :-

(A) 8.505 MeV (B) 8.605 MeV

(C) 9.23 MeV (D) 9.315 MeV

2. When a monochromatic point source of light is at a distance of 0.2 m from a photo-electric cell, the stopping potential and the saturation current are respectively 0.6 volt and 18.0 mA. If the same source is placed 0.6 m away from the photoelectric cell, then :-

(a) The stopping potential will be 0.2 volt
(b) The stopping potential will be 0.6 volt
(c) The saturation current will be 6.0 mA
(d) The saturation current will be 2.0 mA

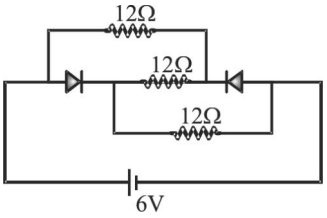
(A) a, c (B) a, d (C) b, c (D) b, d

3. The energy of a photon of wavelength 6600 Å in watt-hour is :-

(A) 3×10^{-19} (B) 1.875

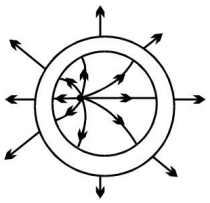
(C) 8.33×10^{-23} (D) 5×10^{-21}

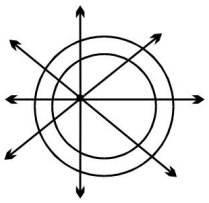
4. Current drawn from the cell in the given circuit :-

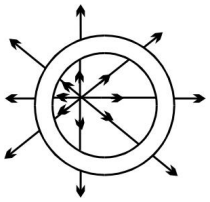


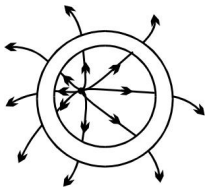
(A) $\frac{1}{6}$ A (B) $\frac{1}{2}$ A (C) $\frac{1}{3}$ A (D) 2A

5. A point charge q is placed at a point inside a hollow conducting sphere. Which of the following electric force pattern is correct ?

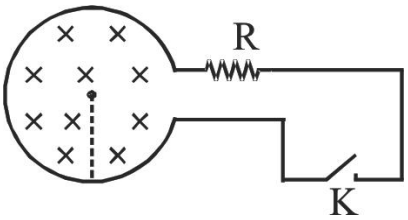
(A) 

(B) 

(C) 

(D) 

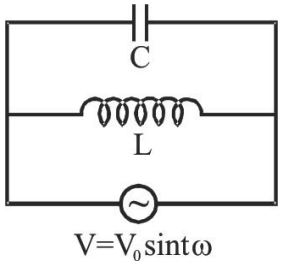
6. Shown in the figure is a circular loop of radius r and resistance R. A variable magnetic field of induction $B = B_0e^{-t}$ is established inside the coil. If the key (K) is closed, the electrical power developed right after closing the switch is equal to :-



(A) $\frac{B_0^2\pi r^2}{R}$ (B) $\frac{B_0 10r^3}{R}$

(C) $\frac{B_0^2\pi^2r^4R}{5}$ (D) $\frac{B_0^2\pi^2r^4}{R}$

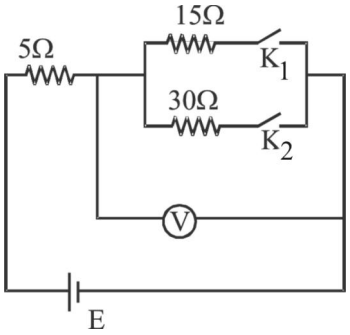
7. For the circuit shown in the figure, the current through the inductor is 0.9 A while the current through the condenser is 0.4A. Hence the current drawn from the generator is:



(A) I = 1.13 A (B) I = 0.9 A

(C) I = 0.5 A (D) I = 0.6 A

8. In the circuit shown in figure reading of voltmeter is V_1 when only K_1 is closed, reading of voltmeter is V_2 when only K_2 is closed and reading of voltmeter is V_3 when both K_1 and K_2 are closed. Then :-



(A) $V_3 > V_2 > V_1$ (B) $V_2 > V_1 > V_3$

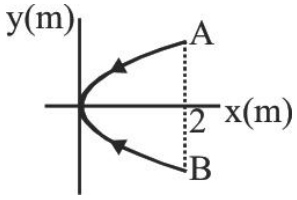
(C) $V_3 > V_1 > V_2$ (D) $V_1 > V_2 > V_3$

9. Condenser A has a capacity of 15μF when it is filled with a medium of dielectric constant 15. Another condenser B has a capacity of 1μF with air between the plates. Both are charged separately by a battery of 100 V. After charging, both are connected in parallel without the battery and the dielectric medium being removed. The common potential now is :-

(A) 400 V (B) 800 V

(C) 1200 V (D) 1600 V

10. A conducting wire bent in the form of a parabola $y^2 = 2x$ carries a current i = 2 A as shown in figure. This wire is placed in a uniform magnetic field $\vec{B} = -4\hat{k}$ tesla. the magnetic force on the wire is (in newton) :-



(A) $-16\hat{i}$ (B) $32\hat{i}$ (C) $-32\hat{i}$ (D) $16\hat{i}$

11. A wave in a string has an amplitude of 2cm. The wave travels in the + ve direction of x axis with a speed of 128 m/sec and it is noted that 5 complete waves fit in 4 m length of the string. The equation describing the wave is :-

(A) y = (0.02) m sin (7.85x – 1005t)

(B) y = (0.02) m sin (7.85x + 1005t)

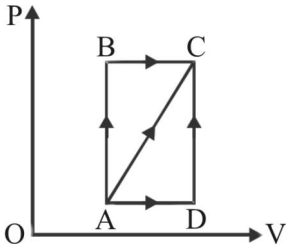
(C) y = (0.02) m sin (15.7x – 2010t)

(D) y = (0.02)m sin (15.7x + 2010t)

12. The number of possible natural oscillations of air column in a pipe closed at one end of length 85 cm whose frequencies lie below 1250 Hz are: (velocity of sound = 340 ms^{-1})

(A) 4 (B) 5 (C) 7 (D) 6

13. A thermodynamic process is shown in the figure. The pressures and volumes corresponding to some points in the figure are:
$P_A = 3 \times 10^4$ Pa, $P_B = 8 \times 10^4$ Pa and
$V_A = 2 \times 10^{-3}$ m^3, $V_D = 5 \times 10^{-3}$ m^3.
In process AB, 600 J of heat is added to the system and in process BC, 200 J of heat is added to the system. The change in internal energy of the system in process AC would be:-



(A) 560 J (B) 800 J

(C) 600 J (D) 640 J

14. The root mean square speed of the molecules of a gas is 1260 m/s. The most probable speed of the molecules is -

(A) 1029 m/s (B) 1161 m/s

(C) 1171 m/s (D) 917 m/s

15. A heavy body of mass 25 kg is to be dragged along a horizontal plane ($\mu = 1/\sqrt{3}$). The least force required is:

(A) 25 kg f (B) 2.5 kg f

(C) 12.5 kg f (D) $25/\sqrt{3}$ kg f

16. From a circle of radius a, an isosceles right angled triangle with the hypotenuse as the diameter of the circle is removed. The distance of the centre of gravity of the remaining portion from the centre of the circle is :-

(A) $\frac{a}{3}$ (B) $\frac{(\pi-1)a}{6}$

(C) $\frac{a}{3(\pi-1)}$ (D) $\frac{a}{3(\pi+1)}$

17. A ball of mass m moving with a velocity v strikes the bob of a pendulum (mass m) at rest. If the collision is perfectly inelastic, the height to which both will rise is given by :-

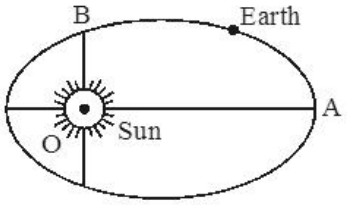
(A) $\frac{v^2}{8g}$ (B) $\frac{v^2}{4g}$ (C) $\frac{v^2}{2g}$ (D) $\frac{v^2}{g}$

18. Two discs have same mass and thickness. Their materials are of densities ρ_1 and ρ_2. The ratio of their moment of inertia about central axis will be-

(A) $\rho_1 : \rho_2$ (B) $\rho_1\rho_2 : 1$

(C) $1 : \rho_1\rho_2$ (D) $\rho_2 : \rho_1$

19. The earth moves around the sun in an elliptical orbit as shown in the figure. The ratio $\frac{OA}{OB} = x$. The ratio of the speeds of the earth at B and at A is :



(A) $\sqrt{x}$ (B) x (C) x^2 (D) $x\sqrt{x}$

20. A large open tank has two holes in the wall. One is a square hole of side L at a depth y from the top and the other is a circular hole of radius R at a depth 4y from the top. When the tank is completely filled with water, the quantities of water flowing out per second from both holes are the same. Then, R is equal to :-

(A) $\frac{L}{\sqrt{2\pi}}$ (B) $2\pi L$

(C) L (D) $\frac{L}{2\sqrt{\pi}}$

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

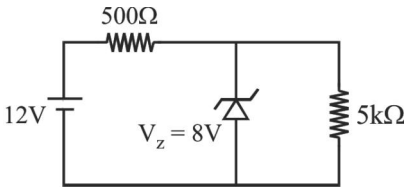
Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

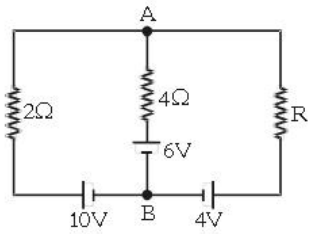
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1. A nuclear reactor generates power by fission of $_{92}U^{235}$ into two equal fragments of $_{46}Pd^{116}$ with the emission of two γ-rays of 5.2 MeV each and three neutrons. The average binding energies per nucleons of $_{92}U^{235}$ and $_{46}Pd^{116}$ are 7.2 MeV and 8.2 MeV respectively. The usable energy released per fission (in MeV) event is :-

2. Power lost in zener diode in the given circuit is 51.2×10^{-x} w, find x :

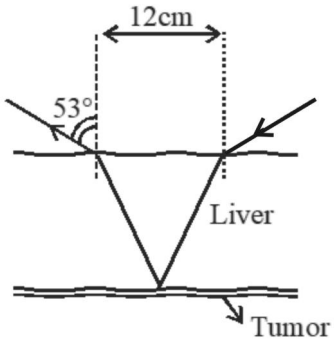


3. In the given circuit as shown in figure the value of R(in Ω) for which current through 4Ω resistor is zero.



4. A conducting rod of 1 m length and 1 kg mass is suspended by two vertical wires through its ends. An external magnetic field of 2T is applied normal to the rod. Now, the current to be passed through the rod so as to make the tension in the wires zero is : (Take g = 10 ms^{-2}) :-

5. A narrow beam of ultrasonic waves reflects from a liver tumer at some depth inside the liver as shown in figure. The speed of the ultrasonic wave is 25% less in the liver than in the surrounding medium. Then depth of the tumor in cm will be:



6. Two coherent sources of equal intensities produce a maximum of 100 units. If the amplitude of one of the sources is reduced by 20%, then the maximum intensity produced will be :

7. A black body at a temperature of 1640 K has the wavelength corresponding to maximum emission equal to 1.75μm. Assuming the moon to be a perfectly black body, the temperature of the moon, if the wavelength corresponding to maximum emission is 14.35μm is (in kelvin)

8. If the volume of a block of metal changes by 0.12% when it is heated through 20°C, the coefficient of linear expansion (in $°C^{-1}$) of the metal is $x \times 10^{-5}$, find the value of x ?

9. The total weight of a piece of wood is 6 kg. In the floating state in water its 1/3 part remains inside the water. On this floating solid, what maximum weight is to be put such that the whole of the piece of wood is to be drowned in the water.

10. A 0.1 kg mass is suspended from a wire of negligible mass. The length of the wire is 1m and its cross–sectional area is 4.9×10^{-7} m^2. If the mass is pulled a little in the vertically downward direction and released, it performs simple harmonic motion of angular frequency 140 rad s^{-1}. If the Young's modulus of the material of the wire is $n \times 10^9$ Nm^{-2}, the value of n is

Topic : Full Syllabus

PART-2 : CHEMISTRY

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.
Zero Marks : 0 If none of the option is selected.
Negative Marks : –1 If wrong option is selected.

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1. Gaseous N_2O_4 dissociates into gaseous NO_2 according to the reaction

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$

At 300 K and 1 atm pressure, the degree of dissociation of N_2O_4 is 0.2. If one mole of N_2O_4 gas is contained in a vessel, then the density of the equilibrium mixture is :

(A) 3.11 g/L (B) 6.22 g/L

(C) 1.56 g/L (D) 4.56 g/L

2. The emf in terms of oxidation potential of half cell given below is:

$$\underset{2\,atm}{Pt\,|\,H_2(g)\,|\,}\underset{(0.02\,M)}{HCl}$$

(A) + 0.1093 V (B) – 0.1143 V

(C) + 0.2286 V (D) – 0.2286 V

3. Electrolysis of hot aqueous solution of NaCl gives $NaClO_4$, i.e., sodium perchlorate,

$NaCl + 4H_2O \longrightarrow NaClO_4 + 4H_2 \uparrow$

How many faraday are required to obtain 1000 g of sodium perchlorate :-

(A) 65.3 (B) 40.3

(C) 18.3 (D) 31.6

4. If 20 mL of 0.1 M NaOH is added to 30 mL of 0.2 M CH_3COOH (pK_a = 4.74), the pH of the resulting solution is :

(A) 4.44 (B) 9.56 (C) 8.96 (D) 9.26

5. If boiling point of an aqueous solution is 100.1°C, what is its freezing point ? Given, enthalpy of fusion and vaporisation of water are 80 cal g^{-1} and 540 cal g^{-1} respectively :-

(A) 0.361°C (B) –0.361°C

(C) –3.61°C (D) None of these

6. Complete combustion of 1.80 g of an oxygen containing compound ($C_xH_yO_z$) gave 2.64 g of CO_2 and 1.08 g of H_2O. The percentage of oxygen in the organic compound is:

(A) 51.63 (B) 63.53

(C) 53.33 (D) 50.33

7. $\Delta S_{Total} = -40$ kJ/mol-K
$\Delta H_{System} = 2000$ kJ/mol
T = 400 K
Find out value of ΔS_{System}.

(A) –35 kJ/mol-K (B) –10 kJ/mol-K

(C) –40 kJ/mol-K (D) –5 kJ/mol-K

8. Consider the following statement which of the following is not true

(A) Eu^{+2} is a strong reducing agent (Z of Eu = 63)

(B) Ce^{+4} is a strong oxidising agent (Z of Ce = 58)

(C) Curium have electronic configuration [Rn] $5f^7\ 6d^1\ 7s^2$ (Z of Cm = 96)

(D) Yb^{+2} is a oxidising agent (Z of Yb = 70)

9. Potassium permanganate acts as an oxidant in neutral, alkaline as well as acidic media. The final products obtained from it in the three conditions are, respectively :-

(A) MnO_4^{2-}, Mn^{3+} and Mn^{2+}

(B) MnO_2, MnO_4^{2-} and Mn^{2+}

(C) MnO_2, MnO_2^{+} and Mn^{3+}

(D) MnO_2, MnO_2 and Mn^{3+}

10. Which of the following bonds has the highest energy

(A) Se–Se (B) Te–Te (C) S–S (D) O–O

11. The ionization potential of nitrogen is more than that of oxygen because of :-

(A) Greater attraction of electrons by the nucleus

(B) Extra stability of the half filled p-orbitals

(C) Smaller size of nitrogen

(D) More penetration effect

12. The option with only amphoteric oxides is :-

(A) Cr_2O_3, NO, SnO, PbO

(B) NO, B_2O_3, PbO, SnO_2

(C) Cr_2O_3, BaO, SnO, SnO_2

(D) ZnO, Al_2O_3,PbO, PbO_2

13. During change of O_2 to O_2^- ion, the electron adds on which one of the following orbitals ?

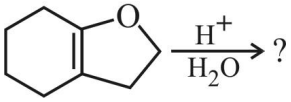
(A) σ* orbital (B) σ orbital

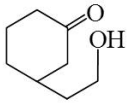
(C) π* orbital (D) π orbital

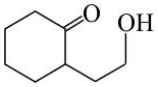
14. Which of the following complex is coloured and diamagnetic -

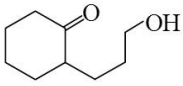
(A) MnO_4^{2-} (B) $[Ni(H_2O)_6]^{2+}$

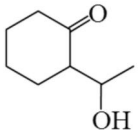
(C) $[CrCl_6]^{3-}$ (D) CrO_4^{2-}

15. 

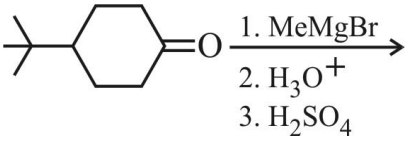
(A) 

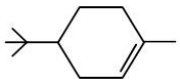
(B) 

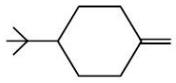
(C) 

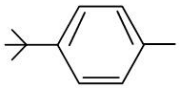
(D) 

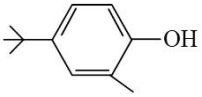
16. Identify the major product in the following reaction.



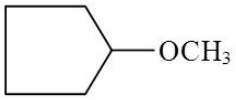
(A) 

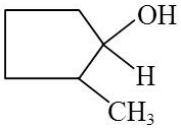
(B) 

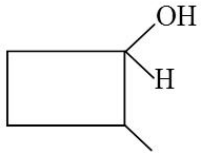
(C) 

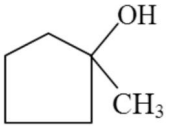
(D) 

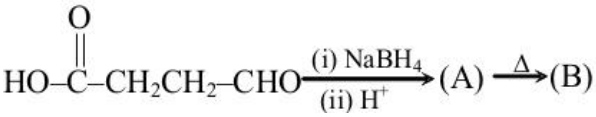
17. An organic compound A (Molecular formula $C_6H_{12}O$) does not change the colour of acidic dichromate solution. Compound A on treatment with H_2SO_4 produces alkene, which on oxidative ozonolysis gives a molecule ($C_6H_{10}O_3$) which gives positive iodoform test find a structure of A

(A) 

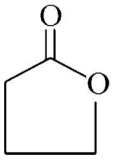
(B) 

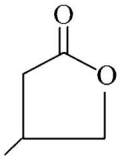
(C) 

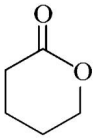
(D) 

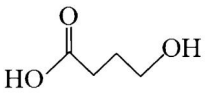
18. 

Structure of B will be :

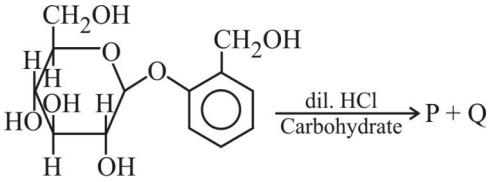
(A) 

(B) 

(C) 

(D) 

19. Salicin (structure given below) is a glycoside, found in the bark of willow tree, used in relieving pain. Observe the following reaction of salicin :



Silicin

The incorrect statement is:

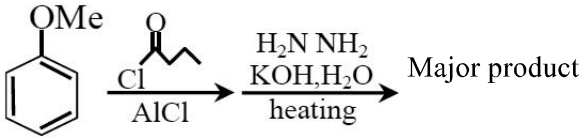
(A) P is D-glucose

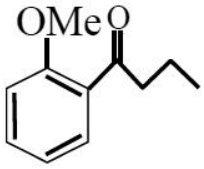
(B) Q is 2-hydroxybenzylalcohol

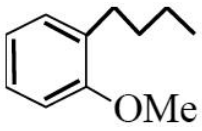
(C) P will show mutaration

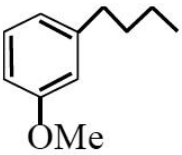
(D) P & Q reduce tollens reagent

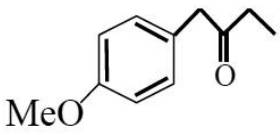
20. What could be the major product for the following reaction :-



(A) 

(B) 

(C) 

(D) 

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

---

1. For a first order reaction, time required for 99.0% completion is 'x' times for the time required for the completion of 90% of the reaction then value of x is ?

2. 3g of acetic acid is added to 250 mL of 0.1 M HCl and the solution made up to 500 mL. To 20 mL of this solution $\frac{1}{2}$ mL of 5 M NaOH is added. The pH of the solution is _____. (Nearest integer)

Given : pK_a of acetic acid = 4.75, molar mass of acetic acid = 60 g/mol, log3 = 0.4771]

Neglect any changes in volume

3. Given the following bond enthalpies :

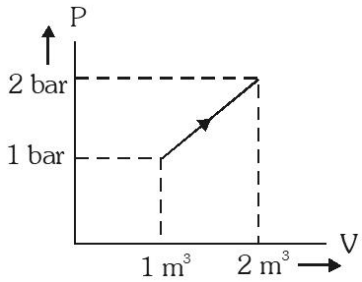
BE(N ≡ N) = 942 kJ/mol ; BE(H – H) = 436 kJ/mol ;

BE(N – N) = 163 kJ/mol ; BE(N – H) = 390 kJ/mol

Determine enthalpy change for the following polymerisation reaction per mole of $N_2(g)$ consumed -

$nN_2(g) + nH_2(g) \longrightarrow -(-NH-NH-)_n-$

4. Find ΔU for the process (in kJ) described by figure if heat supplied during the process is 50 kJ.



5. The value of (n × ℓ × m) for the last electron entered in K is

6. How many of the following ionic compound will have lattice energy less than AlF_3.

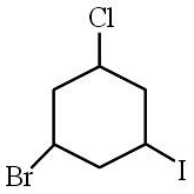
LiF, NaCl, $AlCl_3$, Na_2O, NaBr, KCl, LiI, $BeCl_2$, MgF_2

7. Number of H_2O molecules attached as ligand in $CuSO_4 \cdot 5H_2O$ = x

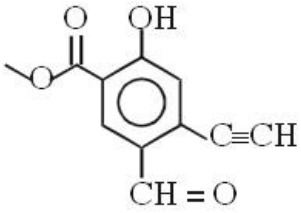
Number of 90° bond angles in XeF_4 = y

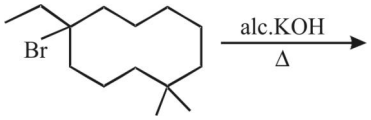
Number of ions in the aqueous solutions of $[Pt(NH_3)_6]Cl_4$ = z find (x + y + z) = ?

8. How many Geometrical Isomers are possible for



9. How many moles of RMgX can be consumed by one mole of following compound when it reacts with excess of RMgX.



10. 

Product mixture.

Number of possible alkenes formed in product mixture.

Topic : Full Syllabus

PART-3 : MATHEMATICS

SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

Full Marks : +4 If correct answer is selected.

Zero Marks : 0 If none of the option is selected.

Negative Marks : –1 If wrong option is selected.

---

1. The value of $\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}+\cos\frac{7\pi}{7}$ is -

(A) 1 (B) -1

(C) 1/2 (D) $-3/2$

2. If $0<\theta<\pi$, then minimum value of $3\sin\theta+\text{cosec}^3\theta$ is:-

(A) 4 (B) 3

(C) 5 (D) 6

3. Mean deviation of first 7 natural no. about their A.M. is :-

(A) 2 (B) $\sqrt{2}$ (C) $\frac{12}{7}$ (D) 0

4. Let $g(x)=f(x)+f(1-x)$ and $f''(x)>0$, $x\in(0,1)$. If g is decreasing in the interval $(0,\alpha)$ and increasing in the interval $(\alpha,1)$, then $\tan^{1}(2\alpha)+\tan^{-1}\left(\frac{1}{\alpha}\right)+\tan^{-1}\left(\frac{\alpha+1}{\alpha}\right)$ is equal to :

(A) $\frac{3\pi}{2}$ (B) π (C) $\frac{5\pi}{4}$ (D) $\frac{3\pi}{4}$

5. Area of the region $\{(x,y) : x^2+(y-2)^2\leqslant 4,\ x^2\geqslant 2y\}$ is :

(A) $2\pi-\frac{16}{3}$ (B) $\pi-\frac{8}{3}$

(C) $\pi+\frac{8}{3}$ (D) $2\pi+\frac{16}{3}$

6. Let $y = y(x)$ be the solution of the differential equations

$$\frac{dy}{dx}+\frac{5}{x(x^5+1)}y=\frac{(x^5+1)^2}{x^7},\ x>0.$$

If $y(1)=2$, then $y(2)$ is equal to

(A) $\frac{637}{128}$ (B) $\frac{679}{128}$

(C) $\frac{693}{128}$ (D) $\frac{697}{128}$

7. Let f and g be two functions defined by

$$f(x)=\begin{cases}x+1, & x<0\\ |x-1|, & x\geqslant 0\end{cases}$$ and

$$g(x)=\begin{cases}x+1, & x<0\\ 1, & x\geqslant 0\end{cases}$$

Then (gof) (x) is

(A) Differentiable everywhere

(B) Continuous everywhere but not differentiable exactly at one point

(C) Not continuous at $x=-1$

(D) Continuous everywhere but not differentiable at $x=1$

8. The domain of the function

$f(x) = \frac{1}{\sqrt{[x]^2 - 3[x] - 10}}$ is (where [x] denotes the greatest integer less than or equal to x)

(A) $(-\infty, -2) \cup (5, \infty)$

(B) $(-\infty, -3] \cup [6, \infty)$

(C) $(-\infty, -2) \cup [6, \infty)$

(D) $(-\infty, -3] \cup (5, \infty)$

9. Assume that each born child is equally likely to be a boy or a girl. If two families have two children each, then the conditional probability that all children are girls given that at least two are girls is :

(A) $\frac{1}{11}$ (B) $\frac{1}{17}$ (C) $\frac{1}{10}$ (D) $\frac{1}{12}$

10. Let z_0 be a root of the quadratic equation, $x^2 + x + 1 = 0$. If $z = 3 + 6iz_0^{81} - 3iz_0^{93}$, then arg z is equal to:

(A) $\frac{\pi}{4}$ (B) $\frac{\pi}{3}$ (C) 0 (D) $\frac{\pi}{6}$

11. If both the roots of the quadratic equation $x^2 - mx + 4 = 0$ are real and distinct and they lie in the interval [1,5], then m lies in the interval:

(A) (4, 5) (B) (3, 4)

(C) (5, 6) (D) (−5, −4)

12. The set of all values of λ for which the system of linear equations.

$x - 2y - 2z = \lambda x$

$x + 2y + z = \lambda y$

$-x - y = \lambda z$

has a non-trivial solution.

(A) contains more than two elements

(B) is a singleton

(C) is an empty set

(D) contains exactly two elements

13. A group of students comprises of 5 boys and n girls. If the number of ways, in which a team of 3 students can randomly be selected from this group such that there is at least one boy and at least one girl in each team, is 1750, then n is equal to :

(A) 25 (B) 28

(C) 27 (D) 24

14. Let S_n denote the sum of the first n terms of an A.P. If $S_4 = 16$ and $S_6 = -48$, then S_{10} is equal to :

(A) -200 (B) -320 (C) 400 (D) -400

15. The remainder, when 7^{103} is divided by 17 is ___.

(A) 12 (B) 13

(C) 7 (D) 5

16. If $\vec{a} = \hat{i} + 2\hat{k}$, $\vec{b} = \hat{i} + \hat{j} + \hat{k}$, $\vec{c} = 7\hat{i} - 3\hat{k} + 4\hat{k}$, $\vec{r} \times \vec{b} + \vec{b} \times \vec{c} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$ then $\vec{r} \cdot \vec{c}$ is equal to :

(A) 34 (B) 12 (C) 36 (D) 30

17. Let $\vec{a} = 4\hat{i} + 3\hat{j}$ and $\vec{b} = 3\hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{c}$ is a vector such that $\vec{c} \cdot \left(\vec{a} \times \vec{b}\right) + 25 = 0$, $\vec{c} \cdot \left(\hat{i} + \hat{j} + \hat{k}\right) = 4$ and projection of $\vec{c}$ on $\vec{a}$ is 1, then the projection of $\vec{c}$ on $\vec{b}$ equals :

(A) $\frac{5}{\sqrt{2}}$ (B) $\frac{1}{5}$ (C) $\frac{1}{\sqrt{2}}$ (D) $\frac{3}{\sqrt{2}}$

18. A straight line cuts off the intercepts OA = a and OB = b on the positive directions of x–axis and y–axis respectively. If the perpendicular from origin O to this line makes an angle of $\frac{\pi}{6}$ with positive direction of y–axis and the area of is ΔOAB, $\frac{98}{3}\sqrt{3}$ then $a^2 - b^2$ is equal to :

(A) $\frac{392}{3}$ (B) 196 (C) $\frac{196}{3}$ (D) 98

19. The shortest distance between the lines $\frac{x-5}{1} = \frac{y-2}{2} = \frac{z-4}{-3}$ and $\frac{x+3}{1} = \frac{y+5}{4} = \frac{z-1}{-5}$ is :

(A) $7\sqrt{3}$ (B) $5\sqrt{3}$

(C) $6\sqrt{3}$ (D) $4\sqrt{3}$

20. Let the ellipse E : $x^2 + 9y^2 = 9$ intersect the positive x- and y-axes at the points A and B respectively Let the major axis of E be a diameter of the circle C. Let the line passing through A and B meet the circle C at the point P. If the area of the triangle which vertices A, P and the origin O is $\frac{m}{n}$, where m and n are coprime, then m – n is equal to

(A) 18 (B) 16 (C) 17 (D) 15

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is a **Numerical Value.**

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is entered.

Zero Marks : 0 If the question is unanswered.

Negative Marks : –1 If wrong answer is entered.

---

1. If $\sec\theta + \tan\theta = \frac{1}{3}$, then value of $16\cot^2\theta - 5\sin\theta$ is :

2. If the domain of the function $f(x) = \sec^{-1}\left(\frac{2x}{5x+3}\right)$ is $[\alpha, \beta) \cup (\gamma, \delta]$ then $|3\alpha + 10(\beta + \gamma) + 21\delta|$ is equal to _____ .

3. The number of relations, on the set $\{1,2,3\}$ containing (1,2) and (2,3), which are reflexive and transitive but not symmetric, is_____

4. If $\int_{0.15}^{0.15} \left| 100x^2 - 1 \right| dx = \frac{k}{3000}$, then k is equal to____.

5. Let $I(x) = \int \sqrt{\frac{x+7}{x}}dx$ and $I(9) = 12 + 7\log_e 7$. If $I(1) = \alpha + 7\log_e\left(1+2\sqrt{2}\right)$, then α^4 is equal to_____.

6. Let [x] be the greatest integer $\leq$ x. Then the number of points in the interval $(-2,1)$, where the function $f(x) = |[x]| + \sqrt{x-[x]}$ is discontinuous, is__.

7. If some three consecutive in the binomial expansion of $(x+1)^n$ is powers of x are in the ratio 2 : 15 : 70, then the average of these three coefficient is :-

8. If a and b are the roots of equation $x^2 - 7x - 1 = 0$, then the value of $\frac{a^{21} + b^{21} + a^{17} + b^{17}}{a^{19} + b^{19}}$ is equal to __ .

9. If P(h,k) be point on the parabola $x = 4y^2$, which is nearest to the point Q(0,33), then the distance of P from the directrix of the parabola $y^2 = 4(x+y)$ is equal to :

10. Let $P(a_1, b_1)$ and $Q(a_2, b_2)$ be two distinct points on a circle with center $C\left(\sqrt{2}, \sqrt{3}\right)$. Let O be the origin and OC be perpendicular to both CP and CQ. If the area of the triangle OCP is $\frac{\sqrt{35}}{2}$, then $a_1^2 + a_2^2 + b_1^2 + b_2^2$ is equal to ___.

ALLEN CAREER INSTITUTE KOTA (RAJASTHAN)

CLASSROOM CONTACT PROGRAMME
(Academic Session : 2023 - 2024)

JEE (Main)
SCORE - II
06-02-2024

JEE(Main) : ENTHUSIAST & LEADER SCORE -II

ANSWER KEY

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	C	B	A	D	C	B	B	B
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	D	A	A	C	C	A	D	B	D
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	200	3	1	5	8	81	200	2	12	4

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	A	B	C	A	D	B	C
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	B	D	C	D	B	A	D	A	D	B
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	2	5	272	-100	0	9	13	4	5	6

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	D	A	C	B	A	C	B	C	A	A
	Q.	11	12	13	14	15	16	17	18	19	20
	A.	A	B	A	B	A	A	A	A	C	C
SECTION-II	Q.	1	2	3	4	5	6	7	8	9	10
	A.	5	24	3	575	64	2	232	51	6	24

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A)

Binding energy of $_{11}N^{23}$ nucleus

$= 0.21 \times 931.5$ MeV

Binding energy per nucleon

$= \dfrac{0.21 \times 931.5}{23} = 8.505$ MeV

2. Ans (D)

v_S is independent of intensity as well as distance while

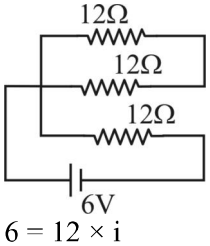
$i_p \propto \dfrac{1}{d^2} \Rightarrow i_p = \dfrac{i_p}{9} = \dfrac{18mA}{9} = 2mA$

3. Ans (C)

$E = \dfrac{hc}{\lambda} = \dfrac{6.6 \times 10^{-34} \times 3 \times 10^8}{6600 \times 10^{-10}} J = 3 \times 10^{-19}$ J

$= \dfrac{3 \times 10^{-19}}{3600} Wh = 8.3 \times 10^{-23}$ Wh

4. Ans (B)



$6 = 12 \times i$

$i = \dfrac{1}{2} A$

6. Ans (D)

Induced emf,

$$E = \frac{d\phi}{dt}$$
$$= \frac{d}{dt}(BA) = A\frac{dB}{dt}$$
$$= \pi r^2 B_0 \frac{d}{dt}(e^{-t})$$
$$= -\pi r^2 B_0 e^{-t}$$

At $t = 0$, $E_0 = B_0\pi r^2$

The electric power developed in the resistor R just at the instant of closing the key is :

$$P = \frac{E_0^2}{R} = \frac{B_0^2\pi^2 r^4}{R}$$

7. Ans (C)

The current drawn by inductor and capacitor will be in opposite phase. Hence, net current by the generator.

$I = I_L - I_C = 0.9 - 0.4 = 0.5A$

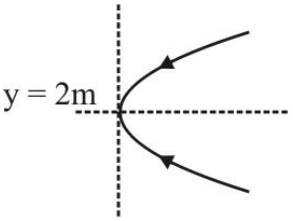
8. Ans (B)

$$V_1 = \frac{15E}{5+15} = \frac{3E}{4} = 0.75E$$
$$V_2 = \frac{30E}{5+35} = \frac{6E}{7} = 0.85E$$
$$V_3 = \frac{10E}{10+5} = \frac{2E}{3} = 0.67E$$
$V_2 > V_1 > V_3$

9. Ans (B)

$$V = \frac{q_1 + q_2}{C_1 + C_2}$$
$$= \frac{15 \times 100 + 1 \times 100}{1+1} = 800 \text{ volt}$$

10. Ans (B)



$\therefore$ Total length = 4m

force = $2 \times 4 \times 4 = 32i$

12. Ans (D)

For COP,

$$n = \frac{xV}{4l} \quad \text{where } x = 1, 3, 5 \ldots.$$
$$n = \frac{x \times 340}{4 \times 85} \times 100$$
$n = 100\ x$

13. Ans (A)

By adjoining graph $W_{AB} = 0$ and

$W_{BC} = 8 \times 10^4[5-2] \times 10^{-3} = 240J$

$\therefore\ W_{AC} = W_{AB} + W_{BC} = 0 + 240 = 240J$

Now,

$\Delta Q_{AC} = \Delta Q_{AB} + \Delta Q_{BC} = 600 + 200 = 800J$

From FLOT $\Delta Q_{AC} = \Delta U_{AC} + \Delta W_{AC}$

$\Rightarrow 800 = \Delta U_{AC} + 240 \Rightarrow \Delta U_{AC} = 560J.$

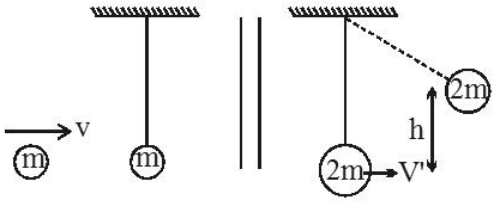
14. Ans (A)

$$V_{rms} = \sqrt{\frac{3RT}{M}} \qquad V_{mp} = \sqrt{\frac{2RT}{M}}$$
$$\frac{V_{rms}}{V_{mp}} = \sqrt{\frac{3}{2}}$$
$$V_{mp} = \sqrt{\frac{2}{3}} \times 1260$$
$$= 1028.78 \text{ m/s.}$$

15. Ans (C)

$$F_{MIN} = \frac{\mu mg}{\sqrt{1+\mu^2}}$$

17. Ans (A)



By conservation of momentum $mv = 2mv'$

$$v^1 = \frac{V}{2}$$

By conservation of mechanical energy $\frac{1}{2} \times (2m)\frac{v^2}{4} = 2mgh$

$$\Rightarrow h = \frac{v^2}{8g}$$

19. Ans (B)

According to law of conservation of angular momentum;

$mv_A \times OA = mv_B \times OB$, $\frac{v_B}{v_A} = \frac{OA}{OB} = x$

PART-1 : PHYSICS

SECTION-II

1. Ans (200)

$_{22}U^{235} \longrightarrow 2(_{46}Pd^{116}) + 3(_0n^1) + 2\ \gamma$-photo(each of 5.2 MeV)

$\therefore$ Required energy

$= 2\ (8.2 \times 116) - (7.2 \times 235) - 2(5.2)$

$= 200$ MeV

2. Ans (3)

Current in zener diode

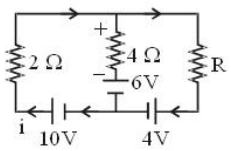
$= \frac{12-8}{500} - \frac{8}{5 \times 10^3} = 6.4$ mA

$\therefore$ Power lost in zener diode

$= VI = 8 \times 6.4$ mW

$= 51.2$ mW

3. Ans (1)



$i = \frac{10-4}{2+R}$, $10 - i \times 2 = 6$

$2 = \frac{10-4}{2+R}$ $i = 2$

$4 + 2R = 6$

$R = 1\Omega$

4. Ans (5)

Magnetic force on rod = $BI\ell$

Weight of the rod = mg

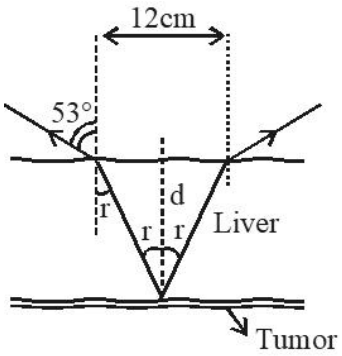
For no tension in wire, $BI\ell = mg$

or $I = \frac{mg}{Bl} = \frac{1 \times 10}{2 \times 1} = 5A$

5. Ans (8)

$\mu = \frac{C}{v} = \frac{C}{3C/4} = \frac{4}{3}$

$1 \sin 53° = \mu \sin r$



$r = 37°$

$\left[\tan 37° = \frac{6}{d}\right]$

$d = 8$ cm

6. Ans (81)

$I_{max} = 4I_0 = 100$

$4I_0 = 100$

$a_0^2 = 25$ or $a_0 = 5$

$a_1 = 5$ and $a_2 = 5$

one amplitude reduced by 20%

$a_1 = 5,\ a_2 = 4$

$I_{max} = (a_1 + a_2)_2 = (5 + 4)^2 = 81$

7. Ans (200)

$\frac{T_2}{T_1} = \frac{\lambda m_1}{\lambda m_2} = \frac{1.75}{14.35} = T_2 = \frac{1.75}{14.35} \times 1640 = 200K$

8. Ans (2)

$\gamma = \frac{\Delta V}{V \Delta T}$

Given $\frac{\Delta V}{V} = 0.12\%$ or 0.0012

$\gamma = \frac{0.0012}{20} = 6 \times 10^{-5}/°C$

Since, $\alpha = \frac{\gamma}{3} = 2 \times 10^{-5}/°C$

9. Ans (12)

upthrust = weight

$\Rightarrow$ Th. = mg

$\Rightarrow v_{in}\rho_w g = mg$

$\Rightarrow \rho_w g = 6g \quad \ldots(A)$

and $(M + 6)\, g = V\rho_w g \quad \ldots(B)$

from equation (A) and (B)

$(M + 6)\, g = 18\, g$

$\Rightarrow M = 12$ kg

10. Ans (4)

$$\omega = \sqrt{\frac{YA}{mL}}$$

$$140 = \sqrt{\frac{n \times 10^9 \times 4.9 \times 10^{-7}}{0.1 \times 1}}$$

$n = 4$

PART-2 : CHEMISTRY

SECTION-I

1. Ans (A)

$N_2O_4 \rightleftharpoons 2NO_2$

1 –

0.8 0.4

$$\bar{m} = \frac{92}{1.2}$$

$$d = \frac{P\bar{m}}{RT} = \frac{1 \times \frac{92}{1.2}}{0.0821 \times 300} = 3.11 \text{gm/L}$$

2. Ans (A)

$$E_{H_2/H^+} = E^-_{H_2/H^+} \frac{0.0591}{2} \log \frac{[H^+]^2}{pH_2} = 0 - \frac{0.0591}{2} \log \frac{(0.02)^2}{2}$$

$$= -\frac{0.0591}{2} \log\left(2 \times 10^{-4}\right) = -\frac{0.0591}{2}[\log 2 + \log 10^{-4}]$$

$$= -\frac{0.0591}{2}[0.3010 - 4.000]$$

$E^=_{H_2/H^+} + 0.1093$ V

3. Ans (A)

$NaCl + 4H_2O \longrightarrow NaClO_4 + 4H_2$

$-1 \qquad +7$

Change in oxidation no. of $NaClO_4$ (V.F.) = 8

Molecular wt. of $NaClO_4$ = 122.5 g/mole

$$\text{Moles of } NaClO_4 = \frac{1000}{122.5} = 8.16$$

No. of Faradays = $8.16 \times 8 \Rightarrow 65.3$

4. Ans (A)

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

Initial milli-moles	6	2	–	–
After reaction	4	–	2	–

$$pH = pK_a + \log \frac{[CH_3COO^-]}{[CH_3COOH]} = 4.44$$

5. Ans (B)

$$\frac{\Delta T_b}{\Delta T_f} = \frac{k_b \times m}{k_f \times m}$$

$$\frac{1}{\Delta T_f} = \frac{0.52 \times m}{1.86 \times m}$$

$$\Delta T_f = \frac{1.86}{0.52} = 3.61°C$$

$\because (F.P)_{solution} = -3.61$ C

6. Ans (C)

$$n_C = n_{CO_2} = \frac{2.64}{44} = 0.06$$

$$n_H = 2 \times n_{H_2O} = \frac{1.08}{18} \times 2 = 0.12$$

$$m_0 = 1.80 - 12 \times \frac{2.64}{44} - \frac{1.08}{18} \times 2$$

$$= 1.80 - 0.72 - 0.12 = 0.96 \text{ gm}$$

$$\% = \frac{0.96}{1.80} \times 100 = 53.33\%$$

7. Ans (A)

$\Delta G = \Delta H - T\Delta S$

$-T\Delta S_{Total} = \Delta H - T\Delta S_{sys.}$

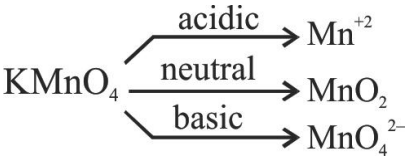
$\Delta S_{sys.} = -35$ kJ/mol-K

8. Ans (D)

$Yb^{+2} \rightarrow Yb^{+3} + e^-$

Yb^{+3} is more stable $\therefore$ Yb^{+2} is reducing agent.

9. Ans (B)



10. Ans (C)

Bond energy

S–S > O–O > Se–Se > Te–Te

11. Ans (B)

'N' has greater IP than 'O' because 'N' has half filled 2p-orbitals.

12. Ans (D)

ZnO, Al_2O_3, PbO and PbO_2 are amphoteric oxides.

13. Ans (C)

Because O_2 has two unpaired e^-s in π ABMO (valance e^-)], so $O_2 \longrightarrow O_2^-$ el^- is entered in ABMO orbital.

14. Ans (D)

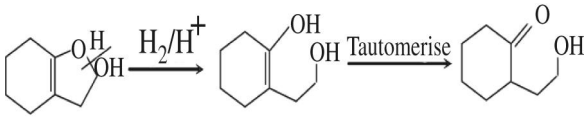
CrO_4^{-2}

Cr^{+6} [Ar] 3d°

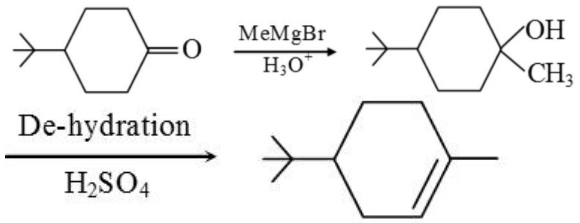
Coloured ⇒ due to charge transfer (L ⟶ m)

diamagnetic ⇒ 3d°

15. Ans (B)

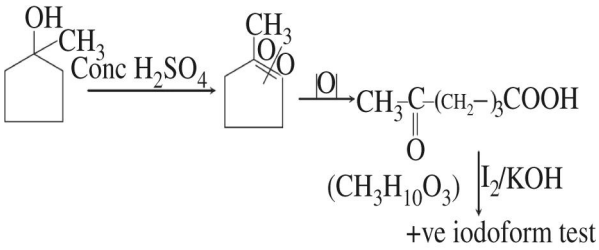


16. Ans (A)

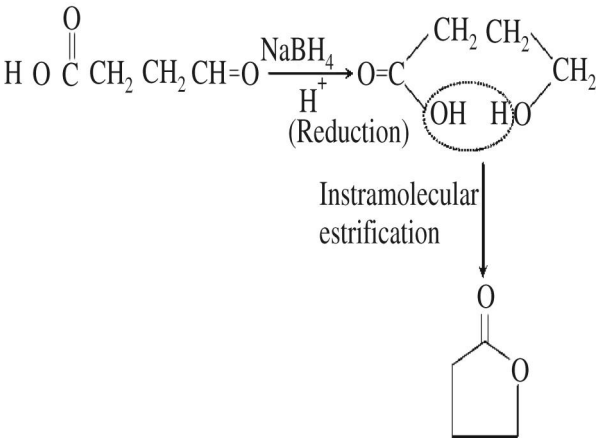


17. Ans (D)

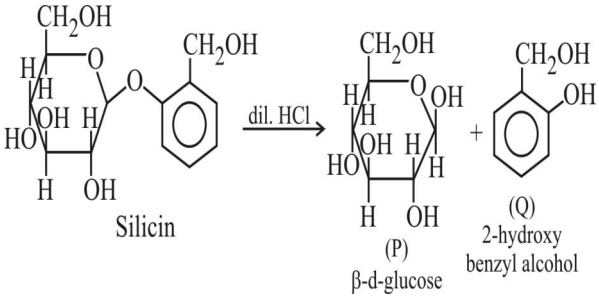
A⟶Must be 3° alcohol because dichromate test do not given by 3° alcohol



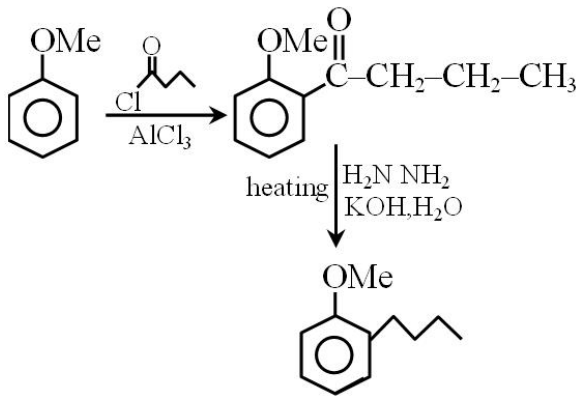
18. Ans (A)



19. Ans (D)



20. Ans (B)



PART-2 : CHEMISTRY

SECTION-II

1. Ans (2)

Apply : $kt = ln\left(\frac{C_0}{C_t}\right)$

2. Ans (5)

3gm Acetic Acid + 250 ml 0.1 M HCl +Water → made to 500 ml solution.

⇒ 500 ml solution has 25 meq of HCl

50 meq of CH_3COOH

∴ 20ml solution has 1 meq of HCl

2 meq of CH_3COOH

We have added 2.5 meq. of NaOH $\left(5M, \frac{1}{2}ml\right)$

Finally , NaOH & HCl are completely consumed and we are left with 0.5 meq of CH_3COOH

and 1.5 meq of CH_3 COONa

$pH = pK_a + \log \frac{1.5}{0.5}$

$= 4.75 + \log 3 = 4.75 + 0.4771$

$= 5.2271$

3. Ans (272)

$nN_2(g) + nH_2(g) \longrightarrow -(-NH-NH-)_n-$

$\Delta H = [n \times BE(N \equiv N) + n \times BE(H-H)$

$- BE(N-N) \times 2n + 2n \times BE(N-H)]$

∵ n = 1

⇒ $\Delta H = [BE(N \equiv N) + BE(H-H) -$

$[BE(N-N) + BE(N-H)] \times 2$

4. Ans (-100)

$W = -\left[10^5_{N/m^2} \times 1\,m^3 + \frac{1}{2} \times 1\,m^3 \times 10^5_{N/m^2}\right]$

$\Delta U = q + w = 50000\,J - 150000\,J = -100\,KJ$

5. Ans (0)

$K[Ar]4s^1$

n = 4,

ℓ = 0,

m = 0

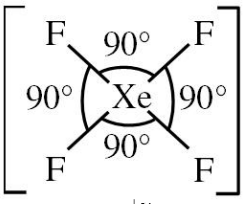
n × ℓ × m = 0

6. Ans (9)

Lattice energy depends on $\frac{q_1q_2}{r}$

7. Ans (13)

$[Cu(H_2O)_4]\,SO_4 \times H_2O \Rightarrow x = 4$



⇒ 90° angles = y = 4

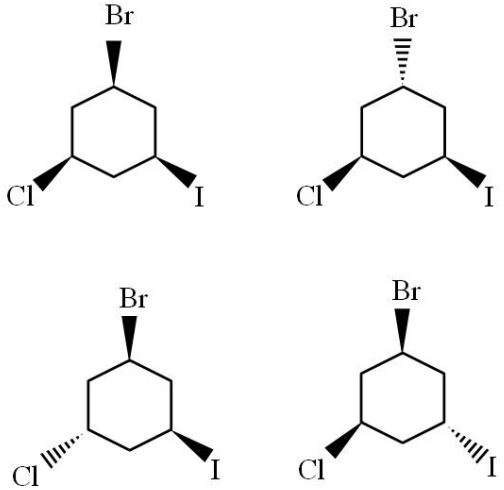
$[Pt(NH_3)_6]^{+y}, 4Cl^-$

(Number of ions = z = 5)

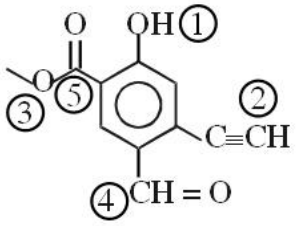
x + y + z

4 + 4 + 5 = 13

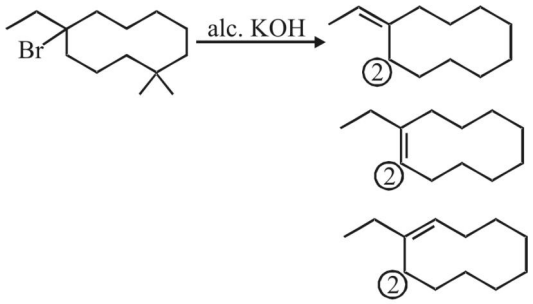
8. Ans (4)



9. Ans (5)



10. Ans (6)



Total (6)

PART-3 : MATHEMATICS

SECTION-I

1. Ans (D)

$$\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}+\cos\frac{7\pi}{7}$$

$$=\cos\frac{2\pi}{7}+\cos\frac{4\pi}{7}+\cos\frac{6\pi}{7}+(-1)$$

$\because \alpha = \frac{2\pi}{7}, \beta = \frac{2\pi}{7}, n = 3$

$$=-1+\frac{\cos\left(\frac{\frac{2\pi}{7}+\frac{6\pi}{7}}{2}\right)\sin\left\{\frac{3}{2}\cdot\frac{2\pi}{7}\right\}}{\sin\left\{\frac{\frac{2\pi}{7}}{2}\right\}}$$

$$=1+\frac{\cos\frac{4\pi}{7}\sin\frac{3\pi}{7}}{\sin\frac{\pi}{7}}=-1+\frac{\cos\frac{4\pi}{7}\sin\left(\pi-\frac{4\pi}{7}\right)}{\sin\frac{\pi}{7}}$$

$$=-1+\frac{2\cos\frac{4\pi}{7}\sin\frac{4\pi}{7}}{2\sin\frac{\pi}{7}}=-1+\frac{\sin\frac{8\pi}{7}}{2\sin\frac{\pi}{7}}=-1+\left(\frac{-1}{2}\right)$$

$$=-\frac{3}{2}$$

2. Ans (A)

Using A.M. $\geq$ G.M.

$$\frac{\sin\theta+\sin\theta+\sin\theta+\cos\sec^3\theta}{4}\geqslant(\sin^3\theta.\cos ec^3\theta)^{1/4}$$

Hence, minimum value of 3 sinq + $\text{cosec}^3\theta$ is 4.

3. Ans (C)

$$\text{Mean deviation}=\frac{\Sigma(x_i-\bar{x})}{N}=\frac{12}{7}$$

4. Ans (B)

$g(x) = f(x) + f(1 - x)$ & $f''(x) > 0, x \in (0, 1)$

$g'(x) = f'(x) - f'(1 - x) = 0$

$\Rightarrow f'(x) = f'(1 - x)$

$x = 1 - x$

$x = \frac{1}{2}$

$g'(x) = 0$

at $x = \frac{1}{2}$

$g''(x) = f''(x) + f''(1 - x) > 0$

g is concave up

hence $a = \frac{1}{2}$

$$\tan^{-1}2\alpha+\tan^{-1}\frac{1}{\alpha}+\tan^{-1}\frac{\alpha+1}{\alpha}$$

$\Rightarrow \tan^{-1}1+\tan^{-1}2+\tan^{-1}3=\pi$

5. Ans (A)

$x^2 + (y-2)^2 \leqslant 2^2$ and $x^2 \geqslant 2y$

Solving circle and parabola simultaneously :

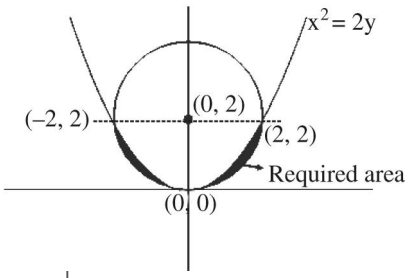
$2y + y^2 - 4y + 4 = 4$

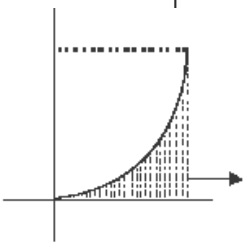
$y^2 - 2y = 0$

$y = 0, 2$

Put $y = 2$ in $x^2 = 2y \longrightarrow x = \pm 2$

$\Rightarrow (2,2)$ and $(-2,2)$





$= 2 \times 2 - \frac{1}{4} \cdot \pi \cdot 2^2 = 4 - \pi$

Required area $= 2\left[\int_0^2 \frac{x^2}{2}dx - (4-\pi)\right]$

$= 2\left[\frac{x^3}{6}\Big|_0^2 - 4 + \pi\right]$

$= 2\left[\frac{4}{3} + \pi - 4\right] = 2\left[\pi - \frac{8}{3}\right] = 2\pi - \frac{16}{3}$

6. Ans (C)

$I.F = e^{\int \frac{5dx}{x\left(x^5+1\right)}} = e^{\int \frac{5x^{-6}dx}{\left(x^{-5}+1\right)}}$

Put, $1 + x^{-5} = t \Rightarrow -5x^{-6}dx = dt$

$\Rightarrow e^{\int \frac{-dt}{t}} = \frac{1}{t} = \frac{x^5}{1+x^5}$

$y \cdot \frac{x^5}{1+x^5} = \int \frac{x^5}{(1+x^5)} \times \frac{\left(1+x^5\right)^2}{x^7}dx$

$= \int x^3dx + \int x^{-2}dx$

$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + c$

Given that : $x = 1 \Rightarrow y = 2$

$2 \cdot \frac{1}{2} = \frac{1}{4} - 1 + c$

$c = \frac{7}{4}$

$y \cdot \frac{x^5}{1+x^5} = \frac{x^4}{4} - \frac{1}{x} + \frac{7}{4}$

Now put, $x = 2$

$y \cdot \left(\frac{32}{33}\right) = \frac{21}{4}$

$y = \frac{693}{128}$

7. Ans (B)

$$f(x) = \begin{cases} x+1, & x < 0 \\ 1-x, & 0 \leqslant x < 1 \\ x-1, & 1 \leqslant x \end{cases}$$

$$g(x) = \begin{cases} x+1, & x < 0 \\ 1, & x \geqslant 0 \end{cases} \Rightarrow$$

$$g(f(x)) = \begin{cases} x+2, & x < -1 \\ 1, & x \geqslant -1 \end{cases}$$

$\therefore g(f(x))$ is continuous everywhere $g(f(x))$ is not differentiable at $x = -1$

Differentiable everywhere else

8. Ans (C)

$[x]^2 - 3[x] - 10 > 0 \Rightarrow [x] < -2 \text{ or } [x] > 5$

9. Ans (A)

$P(B) = P(G) = 1/2$

Required Probability =

$$= \frac{\text{all 4 girls}}{(\text{all 4girls}) + (\text{exactly 3 girls + 1boy}) + (\text{exactly 2girls + 2boys})}$$

$$= \frac{\left(\frac{1}{2}\right)^4}{\left(\frac{1}{2}\right)^4 + {}^4C_3\left(\frac{1}{2}\right)^4 + {}^4C_2\left(\frac{1}{2}\right)^4} = \frac{1}{11}$$

10. Ans (A)

$z_0 = \omega$ or ω^2 (where ω is a non-real cube root of unity)

$z = 3 + 6i(\omega)^{81} - 3i(\omega)^{93}$

$z = 3 + 3i$

$\Rightarrow \arg z = \frac{\pi}{4}$

11. Ans (A)

$x^2 - mx + 4 = 0$

$\alpha, \beta \in [1,5]$



(1) $D > 0 \Rightarrow m^2 - 16 > 0$

$\Rightarrow m \hat{I} (-\infty, -4) \cup (4, \infty)$

(2) $f(1) > 0 \Rightarrow 5 - m > 0$

$\Rightarrow m \in (-\infty, 5]$

(3) $f(5) > 0 \Rightarrow 29 - 5m > 0$

$$\Rightarrow m \in \left(-\infty, \frac{29}{5}\right]$$

(4) $1 < \frac{-b}{2a} < 5 \Rightarrow 1 < \frac{m}{2} < 5 \Rightarrow m \in (2, 10)$

$\Rightarrow m \in (4, 5)$

No option correct : Bonus

* If we consider $\alpha, \beta \in (1,5)$ then option (1) is correct.

12. Ans (B)

$$\begin{vmatrix} \lambda - 1 & 2 & 2 \\ 1 & 2 - \lambda & 1 \\ 1 & 1 & 1 \end{vmatrix} = 0$$

$\Rightarrow (\lambda - 1)^3 = 0 \Rightarrow \lambda = 1$

13. Ans (A)

${}^5C_1 \cdot {}^nC_2 + {}^5C_2 \cdot {}^nC_1 = 1750$

$n^2 + 3n = 700$

$\therefore \ n = 25$

14. Ans (B)

$2\{2a+3d\} = 16$

$3(2a+5d) = -48$

$2a + 3d = 8$

$2a + 5d = -16$

$\boxed{d = -12}$

$S_{10} = 5\{44 - 9 \times 12\}$

$= -320$

15. Ans (A)

$7^{103} = 7 \times 7^{102}$

$= 7(49)^{51}$

$= 7(51 - 2)^{51}$

Remainder :- $7(-2)^{51}$

$\Rightarrow 7(2^3 \cdot (16)^{12})$

$\Rightarrow 56(17 - 1)^{12}$

Remainder = $-56(-1)^{12}$

$= -56 + 68 = 12$

16. Ans (A)

$\vec{r} \times \vec{b} - \vec{c} \times \vec{b} = 0$

$\Rightarrow (\vec{r} - \vec{c}) \times \vec{b} = 0$

$\Rightarrow \vec{r} - \vec{c} = \lambda\vec{b}$

$\Rightarrow \vec{r} = \vec{c} + \lambda\vec{b}$

And given that $\vec{r} \cdot \vec{a} = 0$

$\Rightarrow \left(\vec{c} + \lambda\vec{b}\right) \cdot \vec{a} = 0$

$\Rightarrow \vec{c} \cdot \vec{a} + \lambda\vec{b} \cdot \vec{a} = 0$

$\Rightarrow \lambda = \frac{-\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}}$

Now $\vec{r} \cdot \vec{c} = \left(\vec{c} + \lambda\vec{b}\right) \cdot \vec{c}$

$= \left(\vec{c} - \frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}}\vec{b}\right) \cdot \vec{c} = |\vec{c}|^2 - \left(\frac{\vec{c} \cdot \vec{a}}{\vec{b} \cdot \vec{a}}\right)\left(\vec{b} \cdot \vec{c}\right)$

$= 74 - \left[\frac{15}{3}\right] 8 = 74 - 40 = 34$

17. Ans (A)

$\vec{a} \times \vec{b} = 15\hat{i} - 20\hat{j} - 25\hat{k}$

Let $\vec{c} = x\hat{i} + y\hat{j} + z\hat{k}$

$\Rightarrow 15x - 20y - 25z + 25 = 0$

$\Rightarrow 3x - 4y - 5z = -5$

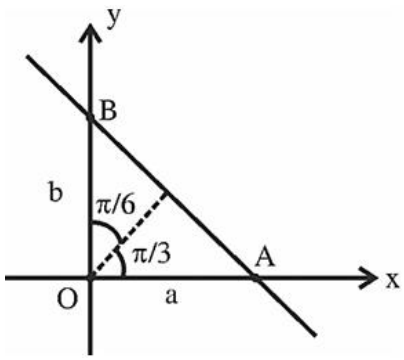
Also $x + y + z = 4$

and $\frac{\vec{c} \cdot \vec{a}}{|\vec{a}|} = 1 \Rightarrow 4x + 3y = 5$

$\Rightarrow \vec{c} = 2\hat{i} - \hat{j} + 3\hat{k}$

Projection of $\vec{c}$ or $\vec{b} = \frac{25}{5\sqrt{2}} = \frac{5}{\sqrt{2}}$

18. Ans (A)



Equation of straight line : $\frac{x}{a} + \frac{y}{b} = 1$

Or $x\cos\frac{\pi}{3} + y\sin\frac{\pi}{3} = p$

$\frac{x}{2} + \frac{y\sqrt{3}}{2} = p$

$\frac{x}{3p} + \frac{y}{2p} = 1$

Comparing both : $a = 2p, b = \frac{2p}{\sqrt{3}}$

Now area of $\Delta OAB = \frac{1}{2} \cdot ab = \frac{98}{3} \cdot \sqrt{3}$

$\frac{1}{2} \cdot 2p \cdot \frac{2p}{\sqrt{3}} = \frac{98}{3} \cdot \sqrt{3}$

$p^2 = 49$

$a^2 - b^2 = 4p^2 - \frac{4p^2}{3} = \frac{2}{3}4p^2$

$= \frac{8}{3} \cdot 49 = \frac{392}{3}$

19. Ans (C)

Shortest distance between two lines

$\frac{x - x_1}{a_1} = \frac{y - y_1}{a_2} = \frac{z - z_1}{a_3}$ &

$\frac{x - x_2}{b_1} = \frac{y - y_2}{b_2} = \frac{z - z_2}{b_3}$ is given as

$$\frac{\begin{vmatrix} x_1 - x_2 & y_1 - y_2 & z_1 - z_2 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}}{\sqrt{(a_1b_3 - a_3b_2)^2 + (a_1b_3 - a_3b_1)^2 + (a_1b_2 - a_2b_1)^2}}$$

$$\frac{\begin{vmatrix} 5-(3) & 2-(-5) & 4-1 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(-10+12)^2 + (-5+3)^2 + (4-2)^2}}$$

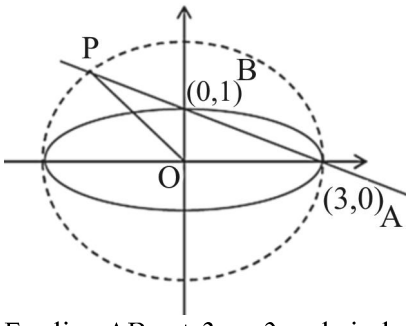
$$\frac{\begin{vmatrix} 8 & 7 & 3 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(2)^2 + (2)^2 + (2)^2}}$$

$= \frac{|8(-10+12) - 7(-5+3) + 3(4-2)|}{\sqrt{4+4+4}}$

$= \frac{16 + 14 + 6}{\sqrt{12}} = \frac{36}{\sqrt{12}} = \frac{36}{2\sqrt{3}}$

$= \frac{18}{\sqrt{3}} = 6\sqrt{3}$

20. Ans (C)



For line AB $x + 3y = 3$ and circle is $x^2 + y^2 = 9$

$(3 - 3y)^2 + y^2 = 9$

$\Rightarrow 10y^2 - 18y = 0$

$\Rightarrow y = 0, \frac{9}{5}$

$\therefore$ Area $= \frac{1}{2} \times 3 \times \frac{9}{5} = \frac{27}{10}$

$m - n = 17$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (5)

$$\sec\theta + \tan\theta = \frac{1}{3}$$
$$\sec\theta - \tan\theta = 3$$

adding $2\sec\theta = \frac{10}{3}$

$$\sec\theta = \frac{5}{3}$$

2. Ans (24)

$f(x) = \sec^{-1}\frac{2x}{5x+3}$

$\left|\frac{2x}{5x+3}\right| \Rightarrow \left|\frac{2x}{5x+3}\right| \geqslant 1 \Rightarrow |2x| \geqslant |5x+3|$

$(2x)^2 - (5x+3)^2 \geq |5x+3|$

$(7x+3)(-3x-3) \geq 0$

$$\frac{- \quad + \quad -}{-1 \qquad -\frac{3}{7}}$$

$\therefore$ domain $\left[-1, \frac{-3}{5}\right) \cup \left(\frac{-3}{5}, \frac{-3}{7}\right]$

$\alpha = -1, \beta = \frac{-3}{5}, \quad \gamma = \frac{-3}{5}, \quad \delta = \frac{-3}{7}$

$3\alpha + 10(\beta + \gamma) + 21\delta = -3$

$\Rightarrow -3 + 10\left(\frac{-6}{5}\right) + \left(\frac{-3}{7}\right)21 = -24$

3. Ans (3)

$A = \{1,2,3\}$

For Reflexive $(1,1)(2,2),(3,3) \in R$

For transitive : $(1,2)$ and $(2,3) \in R \Rightarrow (1,3) \in R$

Not symmetric : $(2,1)$ and $(3,2) \notin R$

$R_1 = \{(1,1),(2,2),(3,3),(1,2),(2,3),(1,3)\}$

$R_2 = \{(1,1),(2,2),(3,3),(1,2),(2,3),(1,3)(2,1)\}$

$R_3 = \{(1,1),(2,2),(3,3),(1,2),(2,3),(1,3)(3,2)\}$

6. Ans (2)

Need to check at doubtful points

discont. at $x \in I$ only

at $x = -1 \Rightarrow f(-1^+) = 1 + 0 = 1$

$\Rightarrow f(-1^-) = 2 + 1 = 3$

at $x = 0 \Rightarrow f(0^+) = 0 + 0 = 0$

$\Rightarrow f(0^-) = 1 + 1 = 2$

at $x = 1 \Rightarrow f(1^+) = 1 + 0 = 1$

$\Rightarrow f(1^-) = 0 + 1 = 1$

discont. at two points

7. Ans (232)

$$\frac{{}^nC_{r-1}}{{}^nC_r} = \frac{2}{15}$$

$$\frac{\frac{n!}{(r-1)!(n-r+1)!}}{\frac{n!}{r!(n-r)!}} = \frac{2}{15}$$

$$\frac{r}{n-r+1} = \frac{2}{15}$$

$15r = 2n - 2r + 2$

$\boxed{17r = 2n + 2}$

$$\frac{{}^nC_r}{{}^nC_{r+1}} = \frac{15}{70}$$

$$\frac{\frac{n!}{r!(n-r)!}}{\frac{n!}{(r+1)!(n-r-1)!}} = \frac{3}{14}$$

$$\frac{r+1}{n-r} = \frac{3}{14}$$

$14r + 14 = 3n - 3r$

$3n - 17r = 14$

$2n - 17r = -2$

$n = 16$

$17r = 34, r = 2$

${}^{16}C_1, {}^{16}C_2, {}^{16}C_3$

$$\frac{{}^{16}C_1 + {}^{16}C_2 + {}^{16}C_3}{3} = \frac{16 + 120 + 560}{3}$$

$$\frac{680 + 16}{3} = \frac{696}{3} = 232$$

8. Ans (51)

$x^2 - 7x - 1 = 0 <^{a}_{b}$

By newton's theorem

$S_{n+2} - 7S_{n+1} - S_n = 0$

$S_{21} - 7S_{20} - S_{19} = 0$

$S_{20} - 7S_{19} - S_{18} = 0$

$S_{19} - 7S_{18} - S_{17} = 0$

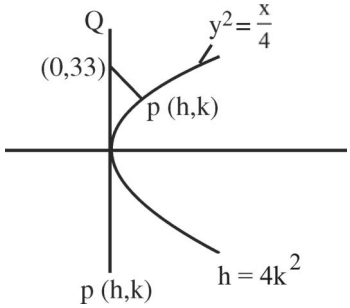
$$\frac{S_{21} + S_{17}}{S_{19}} = \frac{S_{21} + (S_{19} - 7S_{18})}{S_{19}}$$

$$= \frac{S_{21} + S_{19} - 7(S_{20} - 7S_{19})}{S_{19}}$$

$$= \frac{50\,S_{19} + (S_{21} - 7S_{20})}{S_{19}}$$

$$= 51 \cdot \frac{S_{19}}{S_{19}} = \boxed{51}$$

9. Ans (6)



Parabola :

So, $P(4k^2, k)$

$PQ^2 = z = (4k^2 - 0)^2 + (k - 33)^2$

$$\frac{dz}{dk} = 0$$

$2(4k^2) \cdot 8k + 2(k - 33) = 0$

$32k^3 + k - 33 = 0$

$k = 0$

(4,1)

$y^2 = 4(x + y)$

$\Rightarrow\ y^2 - 4y = 4x$

$\Rightarrow\ (y-2)^2 = 4(x + 1)$

Equation of directix :-

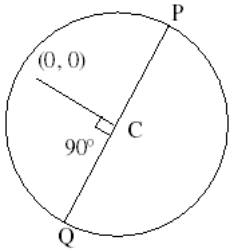
$x + 1 = -1$

$x = -2$

Distance of point = 6

10. Ans (24)

$$\frac{1}{2} \times PC \times \sqrt{5} = \frac{\sqrt{35}}{2};\ PC = \sqrt{7}$$



$a_1^2 + b_1^2 + a_2^2 + b_2^2 = OP^2 + OQ^2$

$= 2(5 + 7) = 24$